

Monic Polynomials

Time Limit: 1.0 seconds

Memory Limit: 256 megabytes

You are given a sequence of N positive integers $A_1, A_2, \dots, A_N$.

For each prefix of length k ($1 \leq k \leq N$), consider the set of k linear monic polynomials in the real polynomial ring $\mathbb{R}[x]$ defined by:

$$f_i(x) = x - A_i \quad \text{for } 1 \leq i \leq k$$

Let $L_k(x)$ be the Least Common Multiple of these polynomials in $\mathbb{R}[x]$:

$$L_k(x) = \text{lcm}(f_1(x), f_2(x), \dots, f_k(x))$$

Since $L_k(x)$ is a monic polynomial of degree d_k , it can be expanded into its standard power basis form:

$$L_k(x) = x^{d_k} - C_{d_k-1} \cdot x^{d_k-1} + C_{d_k-2} \cdot x^{d_k-2} - \dots + (-1)^{d_k} \cdot C_0$$

Your task is to compute the value of the second leading coefficient C_{d_k-1} for every prefix length k from 1 to N .

Input Format

- The first line contains an integer T ($1 \leq T \leq 10^4$) — the number of test cases.
- For each test case:
 - The first line contains an integer N ($1 \leq N \leq 2 \times 10^5$) — the number of elements in the sequence.
 - The second line contains N space-separated integers $A_1, A_2, \dots, A_N$ ($1 \leq A_i \leq 10^9$).

It is guaranteed that the sum of N over all test cases does not exceed 2×10^5 .

Output Format

For each test case, print a single line containing N space-separated integers, where the k_{th} integer represents the coefficient C_{d_k-1} of the polynomial $L_k(x)$.

Sample

Input:

```
2
5
1 2 2 3 1
4
5 5 5 5
```

Output:

```
1 3 3 6 6
5 5 5 5
```

Note

In the first test case:

For $k = 5$: $L_5(x) = \text{lcm}(x - 1, x - 2, x - 2, x - 3, x - 1) = (x - 1)(x - 2)(x - 3) = x^3 - 6x^2 + 11x - 6$. $C_2 = 6$.